Underwater Water-Quality AI Analytics

Anomaly Detection and Next-Step pH Forecasting: Project Report

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1. Abstract

This project builds a reproducible machine-learning pipeline for time-stamped water-quality observations. It has two goals: flag unusual observations (anomaly detection) and predict the next pH value (one-step forecasting). The final system uses a tuned Isolation Forest for anomaly detection and a tuned Random Forest Regressor for forecasting, both frozen before the held-out test set was touched. On the test split the Isolation Forest flags 2.44% of observations (12 of 491), against 4.89% for a Z-score statistical baseline. The Random Forest reaches a forecast MAE of 0.9694 and RMSE of 1.2379 (standardised units), which is 14.4% lower MAE than a persistence baseline.

An additional baseline check performed while preparing this report shows that the forecasting gain needs careful interpretation: the Random Forest improves on a constant "always predict the training mean" model by only 0.4% MAE. The dataset has weak temporal structure, so the result demonstrates the methodology rather than a strong predictive capability. The project also delivers an interactive dashboard (AquaSense AI) with a live simulator, a telemetry replay, a batch processor and a benchmark view.

2. Introduction and Problem Statement

Monitoring water quality over time requires two capabilities. First, observations that depart from normal behaviour must be identified so they can be investigated. Second, near-term values of key parameters should be anticipated. The original problem statement called for time-stamped underwater measurements including temperature, pH, dissolved oxygen and pressure.

The available dataset does not contain those parameters or real timestamps (see Section 3). The project therefore adapts the problem: it demonstrates a complete, leakage-aware methodology on the data that exists, and it states clearly where the results cannot be generalised.

Objectives

- Process timestamped water-quality observations and engineer temporal features.

- Detect anomalous observations with an unsupervised model and compare it with a statistical baseline.
- Forecast the next pH value and compare the model with a naive baseline.
- Evaluate with chronological splits so that no future information leaks into training.
- Package the result as a reusable inference workflow and an interactive dashboard.

3. Dataset

The source is a public water-potability benchmark with 3,276 rows and nine physicochemical measurements plus a binary Potability label (39.0% of rows labelled potable). The Potability label is carried through the pipeline but is not used as a model target.

Feature	Unit	Notes
pH	pH units	491 missing values (15.0%)
Hardness	mg/L	complete
Solids (total dissolved)	mg/L	complete
Chloramines	mg/L	complete
Sulfate	mg/L	781 missing values (23.8%)
Conductivity	µS/cm	complete
Organic carbon	mg/L	complete
Trihalomethanes	µg/L	162 missing values (4.9%)
Turbidity	NTU	complete

Important dataset limitations

- The original data has no real timestamps. Hourly-style timestamps were generated synthetically (2025-01-01 to 2025-05-17) so that temporal methods could be exercised.
- Temperature, dissolved oxygen and pressure, named in the original problem statement, are absent.
- There are no ground-truth anomaly labels, so precision, recall and F1 cannot be computed.

- The rows appear to be independent samples rather than a continuous sensor stream, which limits how much any model can learn from time order (quantified in Section 6.3).

4. Methodology

4.1 Pipeline overview

Raw data → inspection and quality audit → synthetic timestamping → chronological split → imputation and standardisation → feature engineering → anomaly detection and forecasting → tuning on validation data → model freeze → single evaluation on the test split → inference workflow and dashboard.

4.2 Chronological split

Data were split in time order so that models never train on observations later than the ones they are evaluated on.

Split	Rows	Period
Train	2,291	2025-01-01 to 2025-04-07
Validation	491	2025-04-07 to 2025-04-27
Test	491	2025-04-27 to 2025-05-17

The test set was kept untouched until all configurations were frozen.

4.3 Preprocessing and feature engineering

Missing values are filled with a median-type imputer fitted on training data, and features are standardised to z-scores using training statistics. Temporal features are then derived from the timestamp and from pH history:

- Calendar features: hour, day of week, day of month, month.
- pH memory features: previous pH (lag 1), rolling mean of the last three pH values, rolling standard deviation of the last three pH values.
- The forecasting target is the next pH value (ph_next), created by shifting pH one step ahead.

The forecasting model uses 16 inputs (nine sensors, four calendar features, three pH memory features). The anomaly model uses the nine sensor readings only.

4.4 Anomaly detection

Isolation Forest is an unsupervised method that isolates observations with random splits; points that are isolated in few splits receive low scores and are flagged. A Z-score statistical rule served as the baseline. Twenty-seven Isolation Forest configurations were explored (100, 200 and 300 trees; contamination 3%, 5% and 10%; three maximum-sample settings). Because labels are unavailable, the selection criterion was stability of the anomaly rate between training and validation data. The frozen configuration uses contamination 0.03 and max_samples 512, with a model size of about 1.7 MB.

4.5 Forecasting

A Random Forest Regressor predicts the next pH value. A persistence baseline (predict that the next pH equals the current pH) provides the reference. Thirty-six Random Forest configurations were evaluated on validation data; validation MAE ranged narrowly from 0.828 to 0.846 (standardised units). The frozen model uses 300 trees, maximum depth 5, minimum samples to split 5 and minimum samples per leaf 2, with a model size of about 1.1 MB.

4.6 Validation strategy

- Expanding-window chronological validation with three folds (Section 6.4).
- Feature ablation for the anomaly model (Section 6.5).
- Selection on validation data only, model freeze, then one evaluation on the test split.
- A day-25 error and failure analysis documented likely failure conditions without retraining.

5. System Implementation

5.1 Inference workflow

The packaged workflow (src/inference.py) loads a CSV, parses timestamps, builds temporal features, applies the frozen Isolation Forest, applies the frozen Random Forest and writes three outputs per row: anomaly_flag, anomaly_score and predicted_next_ph. Source modules are separated by concern: preprocessing, anomaly detection, forecasting and inference.

5.2 AquaSense AI dashboard

The dashboard (app.py, built with Streamlit and Plotly) provides four workspaces:

- **Interactive Simulator.** Nine sliders with linked numeric inputs, six presets (Dataset Median, Mountain Spring, Municipal Tap, Industrial Runoff, Acid Drainage, Sensor Failure) and a randomiser. Each change runs both frozen models and shows the anomaly verdict and score, the next-step pH, the measured latency, and a z-score bar chart showing which sensors deviate most.
- **Continuous Telemetry Stream.** Replays the 491-observation test split with a configurable window, showing observed pH, the one-step forecast and flagged anomalies.
- **Batch Telemetry Processor.** Accepts a CSV in raw physical units or standardised units, returns flags, scores, forecasts and audit counts, and offers a template and a results download.
- **Model Benchmark and WHO Standards.** Presents the saved validation tables, the chronological folds, the ablation chart, the tuning log and the reference thresholds.

5.3 WHO/EPA reference audit

Alongside the machine-learning models, a deterministic audit checks nine reference limits: pH 6.5 to 8.5, hardness up to 200 mg/L, dissolved solids up to 1,000 mg/L, chloramines up to 4 mg/L, sulfate up to 250 mg/L, conductivity up to 400 µS/cm, organic carbon up to 4 mg/L, trihalomethanes up to 80 µg/L and turbidity up to 5 NTU. These are commonly cited guideline values, several of which are aesthetic or operational rather than health-based, and they are configurable in the code. The audit and the anomaly detector are intentionally independent: a reading can pass every limit and still be flagged because it is unusual relative to the training data, and the reverse is also true.

5.4 Scaler recovery

The models expect standardised inputs, but the original scaler object was not saved in the project. For the dashboard to accept raw physical values, the standardisation parameters were recovered by fitting the relationship between the raw and processed values across all 3,271 uniquely timestamped rows. The fit is exact (residuals near 1e-15). As a check, the dashboard pipeline reproduces all 491 saved anomaly flags in the day-26 inference output and reproduces the published forecast metrics exactly. The recovered parameters are stored in artifacts/day22_final/scaler_recovered.json.

6. Results

All figures below are on the held-out test split. Forecast errors are in standardised units; one unit corresponds to roughly 1.52 pH units, so the Random Forest MAE of 0.9694 is about 1.47 pH units.

6.1 Anomaly detection

Method	Anomaly rate	Processing time (s)	Model size (MB)
Z-score statistical baseline	4.888% (24 of 491)	0.0024	n/a
Tuned Isolation Forest	2.444% (12 of 491)	0.0293	1.665

The Isolation Forest is more selective than the baseline, flagging half as many observations. Without labels this cannot be called more accurate; it shows that the model is stable and conservative (its test rate of 2.44% is close to the 3% contamination target).

6.2 Forecasting against the original baseline

Method	MAE	RMSE
Persistence baseline	1.1327	1.6087
Tuned Random Forest	0.9694	1.2379

MAE falls by 14.4% and RMSE by 23.0% relative to persistence.

6.3 Additional baseline check: constant training mean

Persistence is a strong baseline for smooth time series but a weak one for data with little time dependence. A constant predictor that always outputs the training mean was therefore added and evaluated on the same 487 test rows.

Method	MAE	RMSE
Tuned Random Forest	0.9694	1.2379
Persistence (last value)	1.1327	1.6087
Train-mean constant	0.9728	1.2522

The Random Forest improves on the constant predictor by only 0.4% in MAE and 1.1% in RMSE. The correlation between pH at time t and at time t+1 is 0.17, confirming weak temporal structure. The persistence baseline looks poor mainly because consecutive values are nearly independent, not because the Random Forest has learned a strong pattern.

6.4 Chronological validation

Fold	Training rows	Validation rows	MAE	RMSE
1	1,390	417	0.5759	0.7466
2	1,807	417	0.6849	0.9218
3	2,224	417	0.8351	1.0505

Error grew with each later fold even though more training data was available. This is consistent with the weak temporal signal and with differing variability between periods, and it suggests that additional history alone would not improve this dataset.

6.5 Feature ablation for the anomaly model

Each sensor was removed in turn and the change in validation anomaly rate was recorded. This measures how sensitive the flag rate is to each feature, not how accurate the detector is.

Removed feature	Change in anomaly rate (percentage points)
Chloramines	-1.43
pH	-1.02
Conductivity	-0.81

Removed feature	Change in anomaly rate (percentage points)
Organic carbon	-0.81
Turbidity	-0.61
Sulfate	-0.41
Solids	-0.20
Hardness	0.00
Trihalomethanes	0.00

Chloramines and pH contribute most to the flags; hardness and trihalomethanes contribute none under this configuration.

6.6 Model size and speed

Both models are small (1.7 MB and 1.1 MB) and fast. The saved batch timings for 491 rows are 0.029 s for the Isolation Forest and 0.44 s for the Random Forest. The dashboard measures and displays single-row latency live for both models together.

7. Limitations and Threats to Validity

- **Synthetic time axis.** Timestamps were generated, so temporal results demonstrate the workflow, not performance on real sensor streams.
- **Missing parameters.** Temperature, dissolved oxygen and pressure are absent; conclusions about those variables cannot be drawn.
- **No anomaly labels.** The anomaly rate is descriptive. Precision, recall and F1 are unavailable, and the two methods cannot be ranked by accuracy.
- **Weak temporal structure.** The forecast gain over a constant predictor is marginal (Section 6.3). Claims of forecasting skill should be avoided or heavily qualified.
- **Dataset-relative anomalies.** The Isolation Forest defines normal as typical for this dataset. Pristine water, such as the Mountain Spring preset, can be flagged because typical dataset readings (for example about 22,000 mg/L dissolved solids) are far from drinking-water norms.
- **Reference limits.** The audit thresholds are educational and must be checked against the standard applicable in a given jurisdiction. The dashboard is not a certified water-safety system.

- **Unsaved scaler.** The original scaler was not archived; the recovered parameters are exact for the processed data but the original object should be saved in future runs.
- **Library versions.** The frozen models were serialised with scikit-learn 1.9.1 and should be loaded with a compatible version.

8. Future Work

- Collect or obtain real time-stamped underwater sensor data that includes temperature, dissolved oxygen and pressure.
- Obtain or construct labelled anomaly events to enable precision, recall and F1 evaluation.
- Add stronger forecasting baselines (moving average, linear regression, gradient boosting) and report them alongside the constant predictor.
- Report errors in physical pH units as well as standardised units.
- Save the fitted scaler and imputer together with the models as a single versioned pipeline object.
- Evaluate sequence models only if real data shows meaningful temporal dependence.
- Add per-feature anomaly explanations (for example SHAP) and alert thresholds tied to operational limits.

9. Conclusion

The project delivers a complete, leakage-aware and reproducible pipeline: chronological splitting, validation-only tuning, a frozen model pair, a single test evaluation, an inference workflow and an interactive dashboard. The Isolation Forest provides a stable, conservative anomaly flag rate, and the Random Forest outperforms the persistence baseline by 14.4% MAE. The most important finding is also the most cautious one: because the data has little time dependence, the Random Forest is barely better than predicting the training mean. The work is best presented as a validated methodology and software system whose numerical results will need to be re-established on real sensor data.

Appendix A. Final Model Configuration

Component	Setting
Anomaly model	Isolation Forest, contamination 0.03, max_samples 512, random_state 42
Anomaly features	pH, hardness, solids, chloramines, sulfate, conductivity, organic carbon, trihalomethanes, turbidity
Forecast model	Random Forest Regressor, 300 trees, max depth 5, min samples split 5, min samples leaf 2, random_state 42
Forecast features	the nine sensors, hour, day of week, day, month, ph_lag_1, ph_rolling_mean_3, ph_rolling_std_3
Forecast target	ph_next (next-step pH)
Validation policy	Chronological validation; test set untouched until final evaluation

Appendix B. Repository Layout and How to Run

- src/: preprocessing, anomaly_detection, forecasting, inference, aquaguard (dashboard core).
- artifacts/day22_final/: frozen models, feature configuration, recovered scaler.
- experiments/: tuning logs, ablation and chronological validation, final validation tables, saved inference output.
- notebook/: day-by-day analysis notebooks; docs/: methodology, risk checklist and results notes.
- app.py: dashboard; app_legacy.py: previous sidebar version.

To run: install the packages in requirements.txt, then execute streamlit run app.py. To score a CSV from the command line, run python -m src.inference.